Project Charter – Environment & Climate Analytics

Capstone Project – Project Charter

Group - Members

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Background

Environmental conditions across Atlantic Canada have demonstrated measurable changes over recent decades, driven by a combination of long-term climate change, anthropogenic activity, natural variability, and seasonal meteorological patterns. These changes manifest across interconnected domains, including average temperature trends, precipitation regimes, extreme weather event frequency, greenhouse gas (GHG) emissions, and air quality conditions.

Federal and provincial agencies—including Environment and Climate Change Canada (ECCC) and Climate Change Nova Scotia—publish extensive datasets covering historical climate observations, homogenized long-term climate records, emissions inventories, and air quality indicators. Despite the richness of these datasets, they remain distributed across platforms, formats, temporal frequencies, and reporting frameworks, making integrated environmental analysis challenging.

Assessing how average temperatures and precipitation patterns have evolved over the past 30 years, while simultaneously evaluating extreme weather event frequency, provincial GHG emission changes, and air quality trends, requires reconciling multiple data sources such as station-level climate observations, long-term climate baselines, emissions inventories, and meteorology-influenced AQI measurements. Currently, these factors are typically analyzed in isolation using static tables or siloed reporting tools, with limited support for anomaly detection, trend analysis, correlation exploration, or short-term forecasting across environmental domains.

This project proposes the development of an **Environment & Climate Analytics dashboard** that integrates authoritative public datasets into a unified Business Intelligence (BI) framework, enabling consistent analysis of long-term trends, variability, spatial patterns, and near-term insights for Atlantic Canada, with focused analysis on Nova Scotia.

Problem Statement

Although Canada's environmental datasets are publicly available and scientifically sound, their fragmented structure, inconsistent granularity, and lack of integrated analytical tooling limit their usability for data-driven analysis. Users face difficulty transforming raw environmental data into structured analytical models, applying exploratory and predictive analytics, and communicating insights effectively to non-technical audiences.

From a **Business Intelligence and Data Analytics** perspective, there is a need for an end-to-end analytics solution that consolidates climate, greenhouse gas emissions, and air quality datasets; applies exploratory and predictive methods using tools such as **Python (Pandas), regression techniques, and time-series analysis**; and presents results through **interactive Power BI dashboards**. This project addresses that gap by demonstrating how BI workflows can transform complex environmental data into accessible, decision-ready insights that support analysis, learning, and informed discussion.

Goals

The goals of this capstone project are to:

- Integrate federal environmental datasets into a **unified analytical model** using consistent temporal and spatial dimensions
- Analyze **30-year historical trends (≈1995–2025)** in:
    - Average temperature
    - Precipitation
    - Extreme weather events
- Identify and visualize **temperature and precipitation anomalies** relative to 30-year climate baselines
- Evaluate whether a **measurable trend exists in extreme weather events** (high wind and heavy precipitation) at Nova Scotia weather stations
- Compare **greenhouse gas (GHG) emissions trends** across Canadian provinces and identify **key contributing economic sectors** using data from 2013 to 2023
- Analyze **Air Quality Index (AQI)** trends, seasonal exceedances, and correlations with meteorological conditions
- Develop a **5-year forecast** for average annual temperature at selected Nova Scotia weather stations
- Deliver **interactive, insight-driven Power BI dashboards** that support exploratory and comparative analysis

Scope

The project will deliver an **interactive Environment & Climate Analytics dashboard** focused on **Atlantic Canada**, with enhanced station-level detail for **Nova Scotia**. The scope of analysis integrates:

- Daily climate observations for temperature and precipitation are used to analyze **average annual temperature and total precipitation trends**, **year-over-year variability**, and **anomalies relative to established 30-year climate baselines**. The scope includes indicators of **heavy precipitation events**, **regional comparison across stations and provinces**, and **exploratory short-term temperature forecasting** for selected Nova Scotia weather stations.
- **Annual greenhouse gas (GHG) emissions data** reported by province and economic sector
- **The Air Quality Index (AQI)** focuses on analyzing long-term air quality trends in Canada using data from the National Air Pollution Surveillance (NAPS) Program. The analysis is based on data from **58 active monitoring stations** reporting continuous measurements. Multiple pollutants are assessed, including **$PM_{2.5}$, PM_{10}, O_3, NO_2, NO, NO_x, CO, and SO_2**, using standardized **1-hour, 8-hour, and 24-hour averaging periods**.

The primary analytical periods covered are:

- **Climate data:** approximately **1995–2025**
- **GHG emissions data: 2013–2023**
- **Air quality data: 2000–2024**, based on consistent monitoring availability

The dashboard supports analysis of:

- Average annual temperature trends
- Annual precipitation patterns
- Extreme weather event frequency
- Provincial and sector-level GHG emissions
- AQI behavior, seasonal exceedances, and meteorological correlations
- Short-term temperature forecasting

All data used is sourced exclusively from **public federal datasets**, ensuring transparency, credibility, and reproducibility.

In Scope	Out of Scope
• Integration of historical climate datasets (temperature, precipitation, wind) • Use of **30-year climate baselines** for anomaly analysis • Station-level analysis for Nova Scotia weather stations • Extreme weather event identification and frequency trend assessment • Provincial and sector-level GHG emissions analysis • AQI trend, seasonal exceedance, and correlation analysis • Exploratory data analysis (EDA) using Python (Pandas) • Basic trend modeling (linear / polynomial regression) • Short-term forecasting using time-series methods (e.g., ARIMA) • Export of anomaly and forecast outputs for Power BI • Interactive dashboard development (KPIs, trends, maps, slicers)	• Real-time or streaming environmental data • Long-term climate scenario projections (e.g., CMIP6 future scenarios) • Policy evaluation or emissions reduction recommendation frameworks • Health outcome modeling or impact assessments • Production-grade automated data pipelines or APIs • Custom data collection or sensor deployment

Outcome Expectations

The project will result in a **data-driven analytics dashboard** that provides clear, interpretable insights into **environmental and climate patterns across Atlantic Canada**.

Key outcomes include:

- Clear visualization of long-term climate trends and anomalies
- Evidence-based assessment of extreme weather event frequency
- Identification of provinces and sectors driving GHG emissions change
- Improved understanding of AQI variability and atmospheric influences
- A defensible short-term temperature forecast for Nova Scotia
- Improved environmental data literacy through interactive visual storytelling

Success Criteria

- Public environmental datasets are successfully cleaned, standardized, and integrated into a unified analytical model.
- Core analytical questions related to climate trends, extreme weather, GHG emissions, air quality, and forecasting are clearly addressed.
- Business Intelligence and Data Analytics tools (Pandas, regression/scikit-learn, Power BI) are applied appropriately and effectively.
- Interactive and user-friendly Power BI dashboards with meaningful KPIs and visualizations are delivered.
- Analytical results are validated, with assumptions, limitations, and data definitions clearly documented.
- All project milestones, checkpoints, and final deliverables are completed according to the approved timeline.
- Final outputs meet NSCC academic and professional standards for analysis, documentation, and presentation.

Key Stakeholders

Role	Name
Project Client & Sponsor and Subject Matter Expert	Patrick Dolinger (Faculty, NSCC)
Project Manager	Yashaswi Tiwari (Student, NSCC)
Project Team Members	
• SQL/ETL Analyst	Chisom Njoku (Student, NSCC)
• Data Lead & SQL/ETL Analyst	Danielle Aranha (Student, NSCC)
• Reporting Lead & SQL/ETL Analyst	Michael Okafor (Student, NSCC)
• QA and Documentation	Tracy Odaro (Student, NSCC)

Stakeholder Personas

The Project Sponsor & Subject Matter Expert (Sponsor Patrick)

- **Profile:** Patrick Dolinger is a faculty member at NSCC providing academic oversight and subject matter expertise for the project. He ensures methodological rigor, alignment with program outcomes, and adherence to professional BI standards.

- **Goal:** To guide students in applying sound Business Intelligence and data analytics practices to real-world environmental datasets while ensuring project scope, quality, and academic expectations are met.
- **BI Value:** Patrick uses the dashboards and documented analytical outputs to evaluate methodological correctness, validate assumptions, and ensure insights are defensible, enabling effective knowledge transfer and assessment without direct involvement in data preparation.

Other Stakeholders (Data Consumers): Environmental analysts, planners, policy analysts, students, and educators who use the dashboards and insights for analysis, planning, reporting, and learning purposes.

1. The Policy Planner ("Policy-Maker Paula")

- **Profile:** Government official focused on regional environmental compliance.
- **Goal:** To identify which sectors in Nova Scotia are the primary drivers of GHG emissions to inform new regulations.
- **BI Value:** She uses the **sector-level GHG charts** to get high-level insights for provincial reports without needing to clean raw data herself.

2. The Academic Researcher ("Scholar Sam")

- **Profile:** University student or researcher analyzing long-term climate shifts.
- **Goal:** To validate whether extreme weather events (high wind/rain) have actually increased in frequency over the last 30 years.
- **BI Value:** He relies on the **homogenized climate baselines** and **anomaly visualizations** to prove historical trends with scientifically robust data.

3. The Public Health Advocate ("Analyst Alex")

- **Profile:** Analyst for a non-profit interested in air quality and community well-being.
- **Goal:** To understand the correlation between meteorological patterns and seasonal Air Quality Index (AQI) spikes.
- **BI Value:** Uses **Power BI slicers** to compare seasonal AQI exceedances against temperature and precipitation trends to issue community health warnings.
- **Project Duration**

The project will run from **April 22, 2026 to May 22, 2026**, following a structured academic timeline comprising:

Week / Dates	Milestone / Phase	Key Work & Deliverables
Week 1 (22–24 Apr)	Scoping, Charter & Data Acquisition	Team formation and role assignment. Project charter drafted. Dataset review and acquisition. GitHub repository setup.
Week 2 (27 Apr–1 May)	Data Cleaning & ETL	SQL/Python data cleaning. Power Query ETL. Data dictionary and schema documentation. Checkpoint #1.

Week 3 (4–8 May)	Exploratory Analysis & Predictive Modelling	Pandas EDA. Analytical questions defined. Predictive modelling. Validation and export. Checkpoint #2.
Week 4 (11–15 May)	Power BI Dashboard Build & Iteration	Initial Power BI build. DAX measures and interactivity. QA and stakeholder review. Checkpoint #3.
Week 5 (18–22 May)	Final Delivery & Project Closure	Final dashboard polish. User manual. Presentation rehearsal. Final submission and presentation.

Constraints, Assumptions, and Risks

Constraints

- Reliance on publicly available datasets
- Variation in spatial and temporal granularity across sources
- Limited to free analytics tools (Python, Power BI)

Assumptions

- Public datasets remain accessible and stable
- NSCC-provided tools are sufficient for modeling and visualization

Risks

- Data quality inconsistencies or missing observations
- Time constraints due to fixed academic schedule
- Limited forecasting accuracy due to simplified model

Approval Signatures

Name: **Patrick Dolinger** For Business Intelligence and Data Analytics program (NSCC) **Project Client**	Name: **Patrick Dolinger** Faculty, Technology and Environment, (NSCC) **Project Sponsor**	Name: **Yashaswi Tiwari** Business Intelligence and Data Analytics program Student, NSCC **Project Manager**

Data Sources

This project uses authoritative, publicly available datasets published by federal agencies of the Government of Canada.

Climate, Temperature & Precipitation

- *Environment and Climate Change Canada (ECCC) – Daily Climate Observations Station-level daily climate observations*
    - *https://dd.weather.gc.ca/today/climate/observations/*
    - *https://climate.weather.gc.ca/doc/Technical_Documentation.pdf*
    - *https://climate.weather.gc.ca/*
- *Canadian Centre for Climate Services (CCCS) – Climate Data Access Centralized access point for historical climate datasets, metadata, documentation, and advanced tools supporting baseline and anomaly analysis.*
    - *https://www.canada.ca/en/environment-climate-change/services/climate-change/canadian-centre-climate-services/display-download/climate-data.html*

Greenhouse Gas (GHG) Emissions

- *Canadian Environmental Sustainability Indicators (CESI) –https://www.canada.ca/en/environment-climate-change/services/environmental-indicators/greenhouse-gas-emissions.html*
- *Government of Canada Open Data – National GHG Inventory Machine-readable datasets supporting province-level and sector-level emissions trend analysis for analytical and Business Intelligence use.*

Air Quality Index (AQI)

- *Environment and Climate Change Canada –*
    - *https://www.canada.ca/en/environment-climate-change/services/air-quality-health-index.html*
- *Government of Canada Open Data – National Air Pollution Surveillance (NAPS) Program*
    - *https://open.canada.ca/data/en/dataset/1b36a356-defd-4813-acea-47bc3abd859b*
- *ECCC – NAPS Annual Air Quality Summaries (2000–2024) Annual pollutant summaries used to analyze air quality trends and AQHI behavior across Canada.*
    - *https://data-donnees.az.ec.gc.ca/data/air/monitor/national-air-pollution-surveillance-naps-program/Data-Donnees/2024/ContinuousData-DonneesContinu/AnnualSummaries-SommairesAnnuels?lang=en*